통계열역학 (2013년 1학기) 기말고사 해답

(총 4문제, 100점)

1. **(총 20점, 각 10점)**

(a) 이원자분자의 진동 에너지 준위:

$$\epsilon_n = \left(n+\frac{1}{2}\right)\hbar\omega = \left(n+\frac{1}{2}\right)h\nu = \left(n+\frac{1}{2}\right)hc\bar{\nu} \qquad (n=0,1,2,3,\cdots) \tag{1}$$

위 식에서 $\omega = 2\pi\nu = (k/\mu)^{1/2}$.

$$q_{vib}(T) = \sum_{n=0}^{\infty} e^{-\beta\epsilon_n} = e^{-\beta hc\bar{\nu}/2}\sum_{n=0}^{\infty} e^{-\beta hc\bar{\nu}n} = \frac{e^{-\beta hc\bar{\nu}/2}}{1-e^{-\beta hc\bar{\nu}}} = \frac{e^{-hc\bar{\nu}/(2k_BT)}}{1-e^{-hc\bar{\nu}/(k_BT)}} \tag{2}$$

(b) 이원자분자가 진동 바닥상태에 존재할 확률:

$$p_{n=0}(T) = \frac{e^{-\beta\epsilon_0}}{q_{vib}} \tag{3}$$

이원자분자가 진동 첫번째 들뜬상태에 존재할 확률:

$$p_{n=1}(T) = \frac{e^{-\beta\epsilon_1}}{q_{vib}} \tag{4}$$

$$\frac{p_{n=1}(T)}{p_{n=0}(T)} = \frac{e^{-\beta\epsilon_1}/q_{vib}}{e^{-\beta\epsilon_0}/q_{vib}} = e^{-\beta(\epsilon_1-\epsilon_0)} = e^{-\beta hc\bar{\nu}} = e^{-hc\bar{\nu}/k_BT} = \frac{1}{2}$$
$$\Rightarrow -\frac{hc\bar{\nu}}{k_BT} = -\ln 2 \Rightarrow T = \frac{hc\bar{\nu}}{k_B\ln 2} \tag{5}$$

$\bar{\nu}(\mathrm{I_2}) = 208\ \mathrm{cm^{-1}}$ 이므로 이를 (5) 식에 대입하면,

$$T = \frac{(6.62608\times10^{-34}\ \mathrm{J\ s})(2.998\times10^{10}\ \mathrm{cm\ s^{-1}})(208\ \mathrm{cm^{-1}})}{(1.38066\times10^{-23}\ \mathrm{J\ K^{-1}})(\ln 2)} = 432\ \mathrm{K}$$

2. **(총 20점, 각 10점)**

(a) $\Theta_{rot}(\mathrm{CO}) = 2.77\ \mathrm{K}$ 이므로 300 K에서 $\Theta_{rot}/T = 2.77/300 = 0.00923$ 이다 그러므로 300 K에서 CO 분자가 $J=6$에 존재할 확률은 다음과 같이 주어진다.

$$p_{J=6} = \frac{\Theta_{rot}}{T}(2J+1)e^{-J(J+1)\Theta_{rot}/T} = (0.00923)(2\times6+1)e^{-(6\times7)(0.00923)} = 0.0814$$

(b)

$$\frac{p_{J=6}(T)}{p_{J=0}(T)} = \frac{(2\times6+1)e^{-(6)(7)(2.77)/T}/q_{rot}}{(2\times0+1)e^{-(0)(1)(2.77)/T}/q_{rot}} = 5$$
$$\Rightarrow 13e^{-(6)(7)(2.77)/T} = 5$$
$$\Rightarrow T = -(6)(7)(2.77)/\ln(5/13) = 122\ \mathrm{K} \tag{6}$$

3. **(총 20점, 각 10점)**

(a) 평균에너지 $\langle E\rangle$ 는

$$\begin{aligned}
\langle E\rangle &= k_BT^2\left(\frac{\partial \ln Q}{\partial T}\right)_{N,V} \\
&= k_BT^2\frac{\partial}{\partial T}\ln\left[\frac{1}{N!}\left(\frac{2\pi mk_BT}{h^2}\right)^{3N/2}(V-Nb)^N e^{aN^2/(Vk_BT)}\right] \\
&= k_BT^2\frac{\partial}{\partial T}\left\{-\ln N! + \frac{3N}{2}\ln\left(\frac{2\pi mk_B}{h^2}\right) + \frac{3N}{2}\ln T + N\ln(V-Nb) + \frac{aN^2}{Vk_BT}\right\} \\
&= \frac{3}{2}Nk_BT - \frac{aN^2}{V}
\end{aligned}$$

C_V 는

$$C_V = \left(\frac{\partial\langle E\rangle}{\partial T}\right)_{N,V} = \frac{3}{2}Nk_B$$

(b)

$$\begin{aligned}
P &= -\left(\frac{\partial A}{\partial V}\right)_{N,T} \\
&= k_BT\left(\frac{\partial \ln Q}{\partial V}\right)_{N,T} \\
&= k_BT\frac{\partial}{\partial V}\left\{\ln\left[\frac{1}{N!}\left(\frac{2\pi mk_BT}{h^2}\right)^{3N/2}(V-Nb)^N e^{aN^2/(Vk_BT)}\right]\right\} \\
&= k_BT\frac{\partial}{\partial V}\left\{-\ln N! + \frac{3N}{2}\ln\left(\frac{2\pi mk_BT}{h^2}\right) + N\ln(V-Nb) + \frac{aN^2}{Vk_BT}\right\} \\
&= k_BT\left\{\frac{N}{V-Nb} - \frac{aN^2}{V^2k_BT}\right\} \\
&= \frac{Nk_BT}{V-Nb} - \frac{aN^2}{V^2}
\end{aligned}$$

4. **(총 40점, 각 10점)**

(a) 가로, 세로 길이가 각각 L 인 표면에 흡착되어 있는 단원자분자의 병진 운동에너지는 다음과 같이 주어진다.

$$\epsilon_{n_xn_yn_z} = \frac{h^2}{8maL^2}(n_x^2+n_y^2) \qquad (n_x, n_y = 1, 2, \cdots) \tag{7}$$

그러므로 병진 분자분배함수 $q(S,T)$ 는 다음과 같이 나타 낼 수 있다.

$$
\begin{aligned}
q_{trans}(S,T) &= \sum_{n_x=1}^{\infty}\sum_{n_y=1}^{\infty}\exp\left[-\frac{\beta h^2}{8mL^2}(n_x^2+n_y^2)\right], && \text{(8a)}\\
&= \sum_{n_x=1}^{\infty}\exp\left(-\frac{\beta h^2 n_x^2}{8mL^2}\right)\sum_{n_y=1}^{\infty}\exp\left(-\frac{\beta h^2 n_y^2}{8mL^2}\right)\\
&= \left[\sum_{n=1}^{\infty}\exp\left(-\frac{\beta h^2 n^2}{8mL^2}\right)\right]^2,\\
&\approx \left[\int_0^{\infty}\exp\left(-\frac{\beta h^2 x^2}{8mL^2}\right)dx\right]^2,\\
&= \left(\frac{\pi 8mL^2}{4\beta h^2}\right)\\
&= \left(\frac{2\pi m}{\beta h^2}\right)S,\\
&= \left(\frac{2\pi m k_B T}{h^2}\right)S. && \text{(8b)}
\end{aligned}
$$

(b) 단원자분자들의 구별불가능이므로,

$$
Q(N,S,T) = \frac{q^N}{N!} = \frac{1}{N!}\left(\frac{2\pi m k_B T}{h^2}\right)^N S^N. \qquad (9)
$$

(c) 평균에너지 $\langle E\rangle$ 는

$$
\begin{aligned}
\langle E\rangle &= k_B T^2\left(\frac{\partial \ln Q}{\partial T}\right)_{N,S}\\
&= k_B T^2\frac{\partial}{\partial T}\ln\left[\frac{1}{N!}\left(\frac{2\pi m k_B T}{h^2}\right)^N S^N\right]\\
&= k_B T^2\frac{\partial}{\partial T}\left\{-\ln N! + N\ln\left(\frac{2\pi m k_B}{h^2}\right) + N\ln T + N\ln S\right\}\\
&= N k_B T
\end{aligned}
$$

C_V 는

$$
C_V = \left(\frac{\partial\langle E\rangle}{\partial T}\right)_{N,V} = N k_B
$$

(d)

$$
\begin{aligned}
\phi &= -\left(\frac{\partial A}{\partial S}\right)_{N,T} \\
&= k_B T \left(\frac{\partial \ln Q}{\partial S}\right)_{N,T} \\
&= k_B T \frac{\partial}{\partial S}\left\{ \ln\left[\frac{1}{N!}\left(\frac{2\pi m k_B T}{h^2}\right)^N S^N\right]\right\} \\
&= k_B T \frac{\partial}{\partial S}\left\{ -\ln N! + N \ln\left(\frac{2\pi m k_B T}{h^2}\right) + N \ln S\right\} \\
&= \frac{N k_B T}{S}
\end{aligned}
$$